

10.8.2026Kratfrequencydistributions

Set up

```
library(ggplot2)
library(brms)
```

Loading required package: Rcpp

Loading 'brms' package (version 2.23.0). Useful instructions can be found by typing `help('brms')`. A more detailed introduction to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
library(ggeffects)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v lubridate  1.9.4      v tibble     3.3.1
v purrr      1.2.0      v tidyr      1.3.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
library(here)
```

here() starts at C:/Users/Rwnchr/Documents/OFFICIALKRATBUSINESS/Krat-Senior-thesis-2026-27

```
library(dplyr)
library(ggtext)

#Load in clean data set
df.hlo <- read.csv("10.8.2026KratCleanDataRMC.csv") #w/o head length outlier 36 m
#Load in clean data set with all values
df <- read.csv("RC2026_KRAT_DATA.08.01.csv")
```

Goal

Create frequency distributions for general sizes of each species.

Create frequency distributions for each species over the different seasons of the year.

Our data demographics

36 heermanni, 24 venustus

```
#Remove rows with the most NAs in continuous, quantitative variables or where most measurements are NA
df <- df[-c(6, 10, 11, 12, 13, 14, 15, 16, 19, 20, 21, 22, 23, 24, 25, 34, 36, 37, 38, 39, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100)]

#Show number of each species included in analysis: 36 heermanni, 24 venustus
table(df$Species)
```

```
Dipodomys heermanni  Dipodomys venustus
                 36                24
```

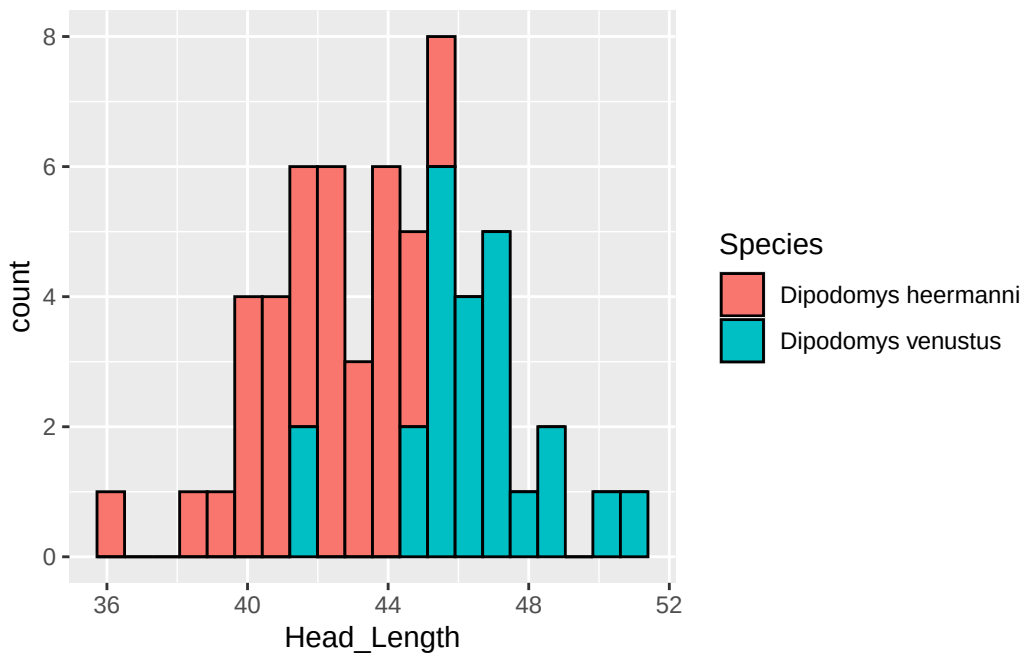
General size frequency distributions for Head Length

With all data points

Option here between a histogram and a density distribution. I like the density one better for looks, but probably tells you less if you don't know about the data for counts and the actual frequency.

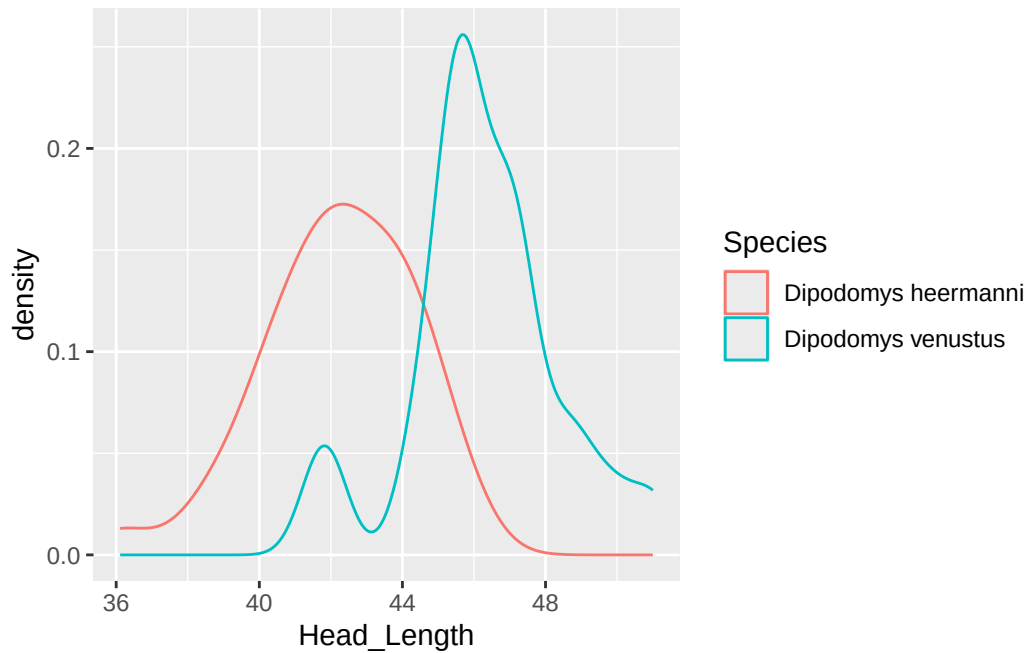
```
#Create a histogram of the head length, separating by species
df %>%
  ggplot(aes(x= Head_Length, fill = Species))+
  geom_histogram(bins = 20, color = "black")
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_bin()`).



```
#Create a density, smooth line frequency distribution, separating by species
df %>%
  ggplot(aes(x= Head_Length, color = Species))+
  geom_density()
```

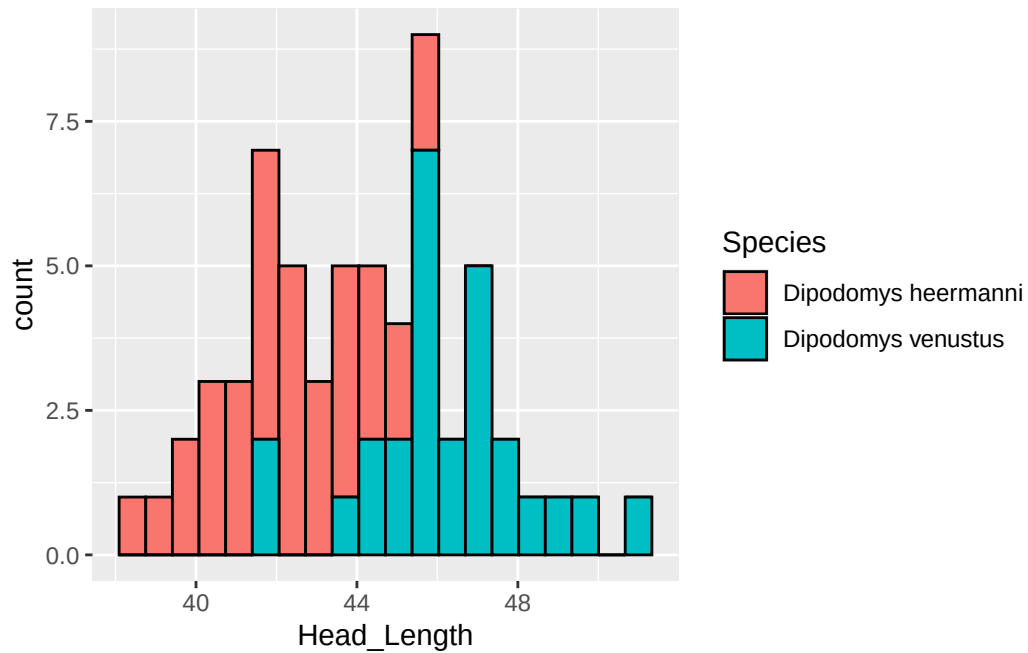
Warning: Removed 1 row containing non-finite outside the scale range (`stat\_density()`).



Without head length outlier (< 37 mm)

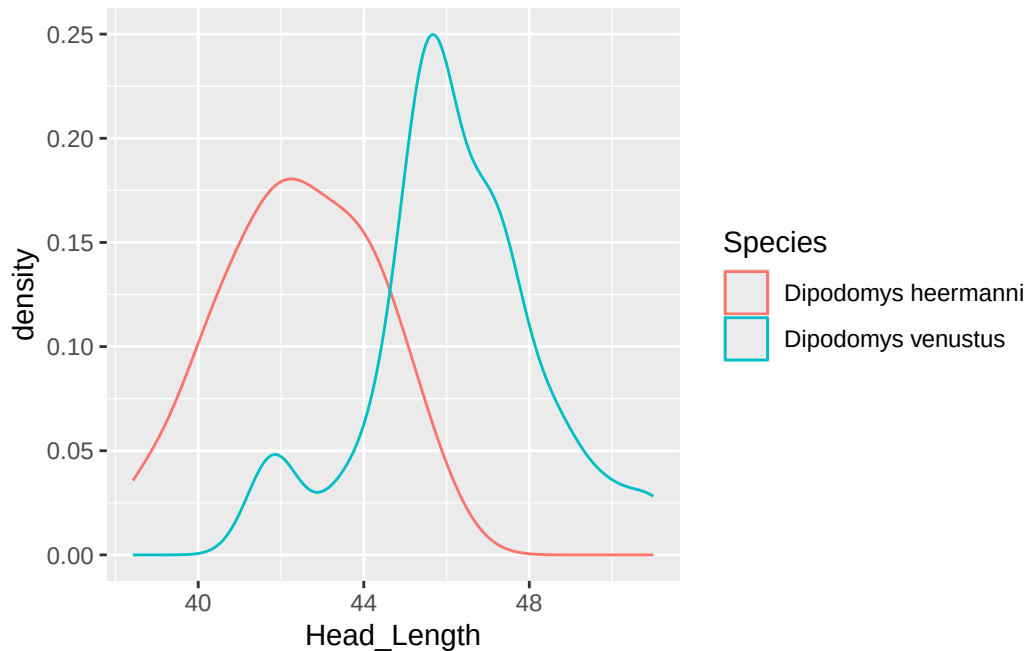
```
#Create a histogram of the head length, separating by species without the outlier
df.hlo %>%
  ggplot(aes(x= Head_Length, fill = Species))+
  geom_histogram(bins = 20, color = "black")
```

Warning: Removed 37 rows containing non-finite outside the scale range
(`stat\_bin()`).



```
#Create a density, smooth line frequency distribution, separating by species without the out.
df.hlo %>%
  ggplot(aes(x= Head_Length, color = Species))+
  geom_density()
```

Warning: Removed 37 rows containing non-finite outside the scale range
(`stat\_density()`).



Tail is left out of the trait distribution when the head length value < 37 mm is not included. Does this mean it's actually an outlier?

Spring data showing the outlier

The more data collected over the summer filled in the gaps of the distribution that had been missing in this spring data analysis. This head length value < 37 may not be an outlier after all??

Observation demographics of old data: 27 heermanni, 14 venustus

```
#Read in old data from the Spring, May 2026
df.old <- read.csv("final_krat_data_2026.csv")

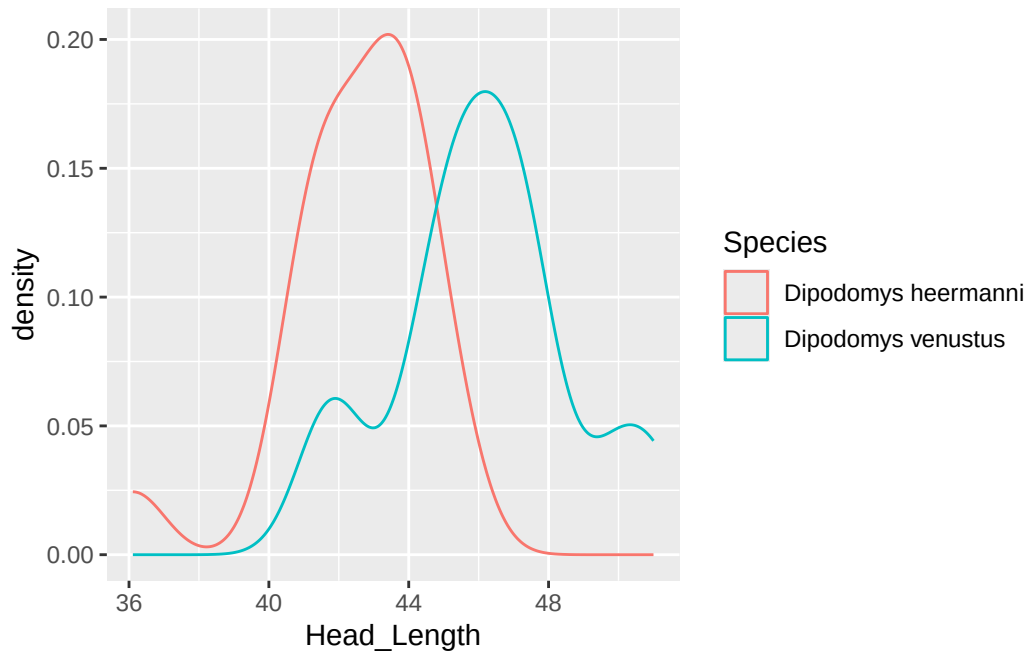
#Remove rows where most data was not taken (recaps mostly)
df.old <- df.old[-c(6, 10, 11, 12, 13, 14, 15, 16, 19, 20, 21, 22, 23, 24, 25, 34, 36, 37, 38,

#Have 27 heermanni, 14 venustus in this analysis from the spring before summer data was taken
table(df.old$Species)
```

<i>Dipodomys heermanni</i>	<i>Dipodomys venustus</i>
27	14

```
#Create a density, smooth line frequency distribution, separating by species
df.old %>%
  ggplot(aes(x= Head_Length, color = Species))+
  geom_density()
```

Warning: Removed 9 rows containing non-finite outside the scale range (`stat\_density()`).



Interpretation: The outlier (< 37mm) in head length absolutely looks like one based on the old data with the lower sample size, but when more data was added the distribution became more normal.

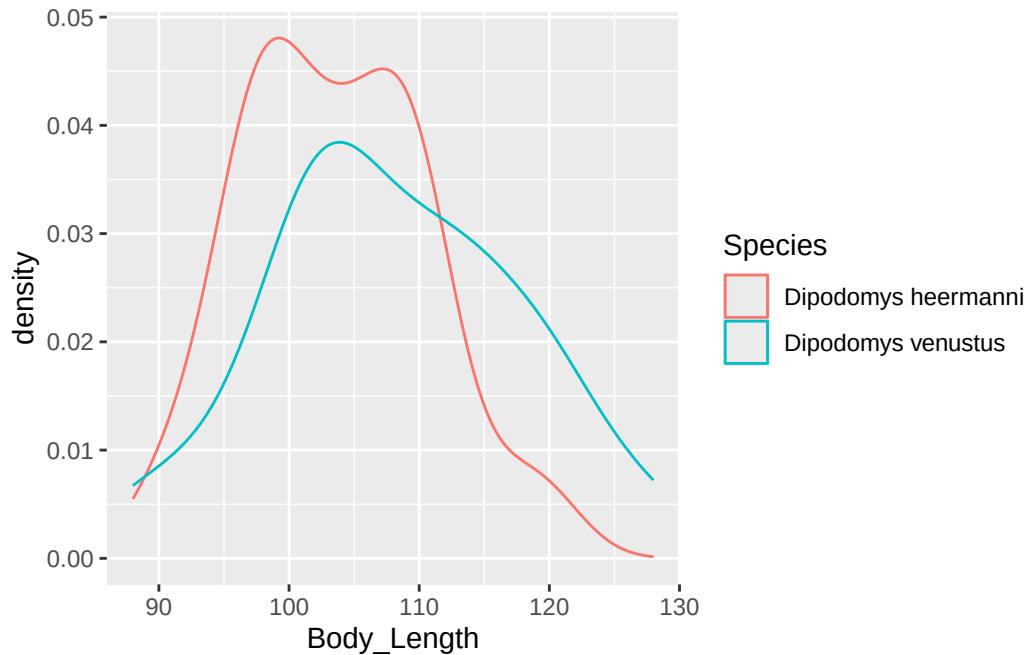
General size frequency distribution for Body Length

With all data points

This graph has the frequency distribution for body length of both species, showing they are not very different in body length, but in the past graphs the head length has differed a lot more. Maybe body length is a better indicator of general size despite it being more coarse??

```
#Create a density, smooth line frequency distribution, separating by species
df %>%
  ggplot(aes(x= Body_Length, color = Species))+
  geom_density()
```

Warning: Removed 1 row containing non-finite outside the scale range
(`stat\_density()`).



Size over the seasons

Observations of both species per season:

Fall: 23, Spring: 3, Summer 33, Winter 1

Both species across the seasons

```
#make a new column for season
df$Season <- NA
head(df$Season) # it worked!
```

```
[1] NA NA NA NA NA NA
```

```
#sort observations by date -- choose what counts as each season
head(df$Date)
```

```
[1] "2025-11-08T16:01:37Z" "2025-11-08T16:20:10Z" "2025-11-08T17:09:58Z"
[4] "2025-11-08T17:19:23Z" "2025-11-09T16:15:25Z" "2025-11-09T18:08:03Z"
```

```
# Extract month and assign season
df <- df %>%
  mutate(
    Month = month(Date),
    Season = case_when(
      Month %in% c(12, 1, 2) ~ "Winter",
      Month %in% c(3, 4, 5) ~ "Spring",
      Month %in% c(6, 7, 8) ~ "Summer",
      Month %in% c(9, 10, 11) ~ "Fall"
    )
  )
head(df$Season) # check that it worked
```

```
[1] "Fall" "Fall" "Fall" "Fall" "Fall" "Fall"
```

```
table(df$Season) #Fall 23, Spring 3, Summer 33, Winter 1
```

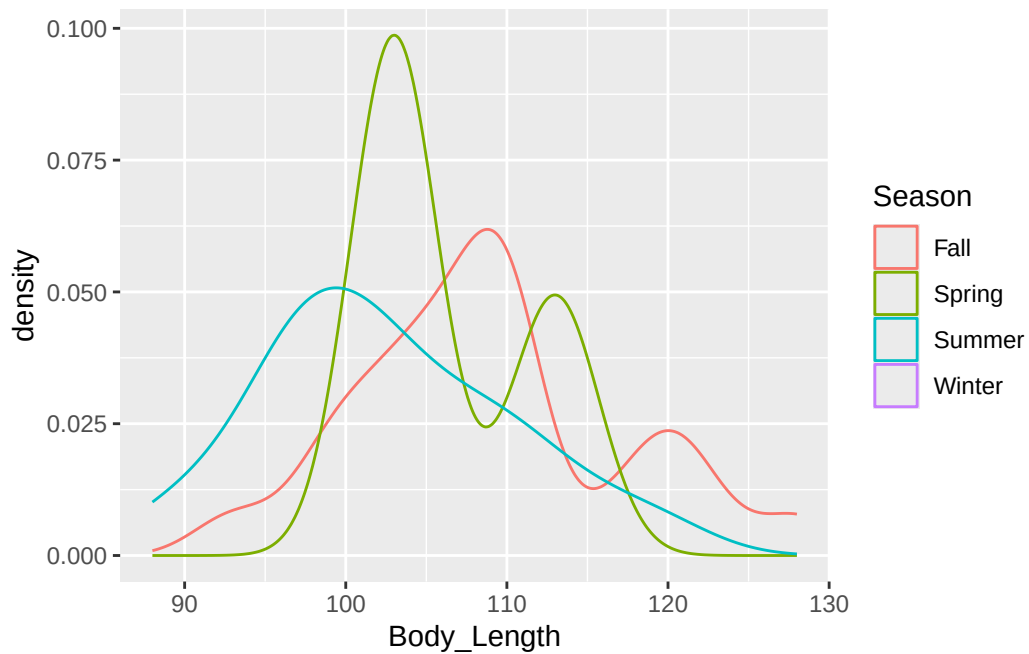
Fall	Spring	Summer	Winter
23	3	33	1

```
#graph with color = season, of all kangaroo rats both species
#Create a density, smooth line frequency distribution, separating by season and including both
df %>%
  ggplot(aes(x= Body_Length, color =Season))+
  geom_density()
```

Warning: Removed 1 row containing non-finite outside the scale range (``stat_density()``).

Warning: Groups with fewer than two data points have been dropped.

Warning: Removed 1 row containing missing values or values outside the scale range (``geom_density()``).



Heermanni across the seasons

For just heermanni observations per season:

Fall: 14

Summer: 22

Winter: 0

Spring: 0

```
#Filter for only D. heermanni  
head(df$Species)
```

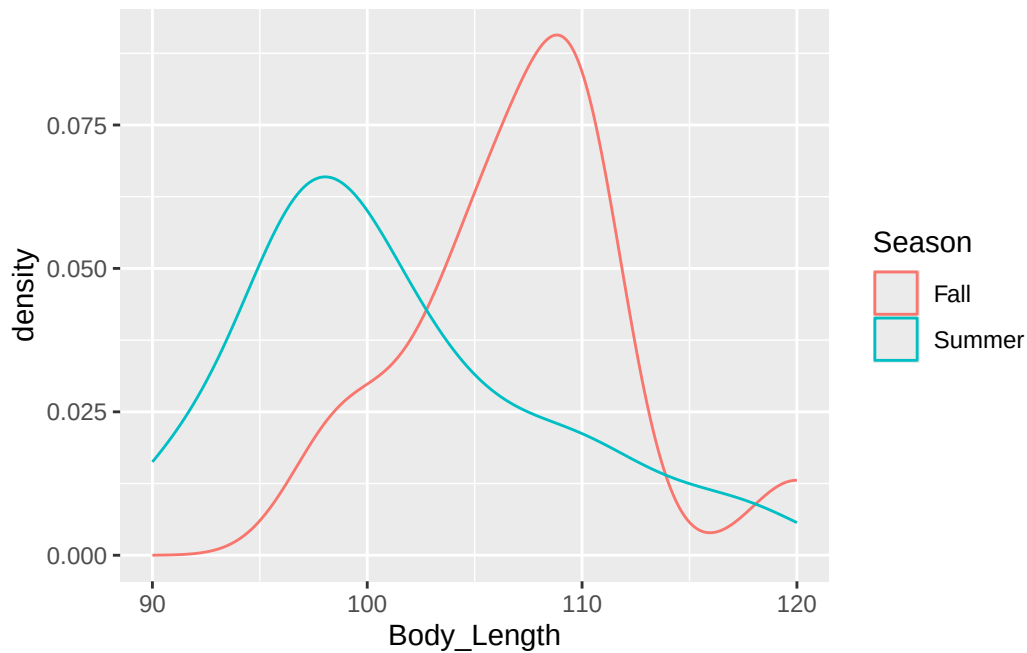
```
[1] "Dipodomys venustus" "Dipodomys venustus" "Dipodomys venustus"  
[4] "Dipodomys venustus" "Dipodomys venustus" "Dipodomys venustus"
```

```
dfhe <- df |> filter(Species == "Dipodomys heermanni")

#Demographics 14 fall, 22 summer
table(dfhe$Season)
```

```
Fall Summer
  14    22
```

```
#Create a density, smooth line frequency distribution, separating by season
dfhe %>%
  ggplot(aes(x= Body_Length, color =Season))+
  geom_density()
```



This seasonal graph of the heermanni is super interesting and might have enough of a sample size to be able to say something?!

Venustus over the seasons

For just venustus observations per season:

Fall: 9

Spring: 3

Summer: 11

Winter: 1

```
#Filter for only D. heermanni  
head(df$Species)
```

```
[1] "Dipodomys venustus" "Dipodomys venustus" "Dipodomys venustus"  
[4] "Dipodomys venustus" "Dipodomys venustus" "Dipodomys venustus"
```

```
dfve <- df |> filter(Species == "Dipodomys venustus")  
  
#Demographics 14 fall, 22 summer  
table(dfve$Season)
```

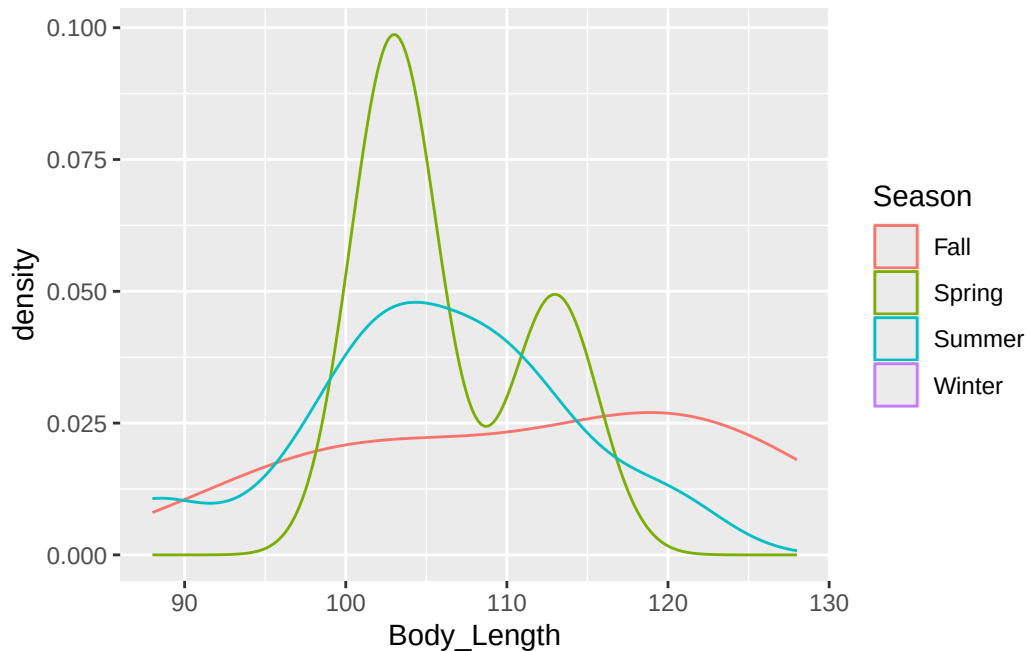
Fall	Spring	Summer	Winter
9	3	11	1

```
#Create a density, smooth line frequency distribution, separating by season  
dfve %>%  
  ggplot(aes(x= Body_Length, color =Season))+  
  geom_density()
```

Warning: Removed 1 row containing non-finite outside the scale range
(`stat\_density()`).

Warning: Groups with fewer than two data points have been dropped.

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_density()`).



This graph here is not based on enough data to mean much, but we don't see the same split in size over the seasons as with the heermanni. This could be due mostly to the low sample size.

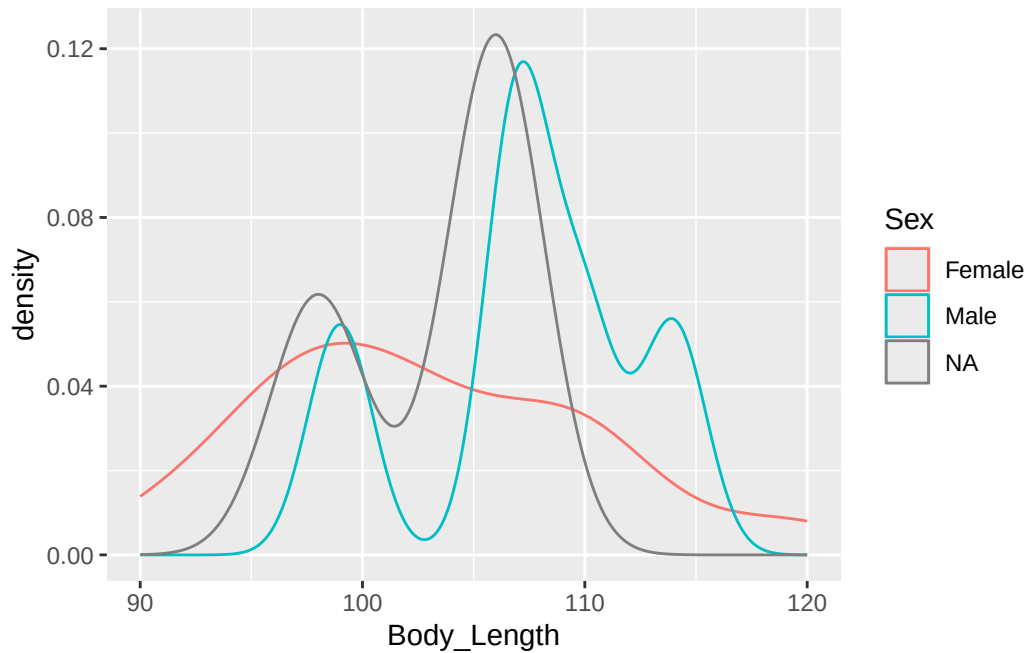
Seasonal sexual dimorphism?

Dipodomys heermanni

Observed 28 Female and 5 Male heermanni.

Females seem to be more abundant than males in both species or maybe they are out foraging more often due to greater metabolic needs (Lactating, pregnant, etc) and therefore get caught in the traps more often??

```
#Create a density, smooth line frequency distribution, of just the heermanni separating by sex
dfhe %>%
  ggplot(aes(x= Body_Length, color =Sex))+
  geom_density()
```

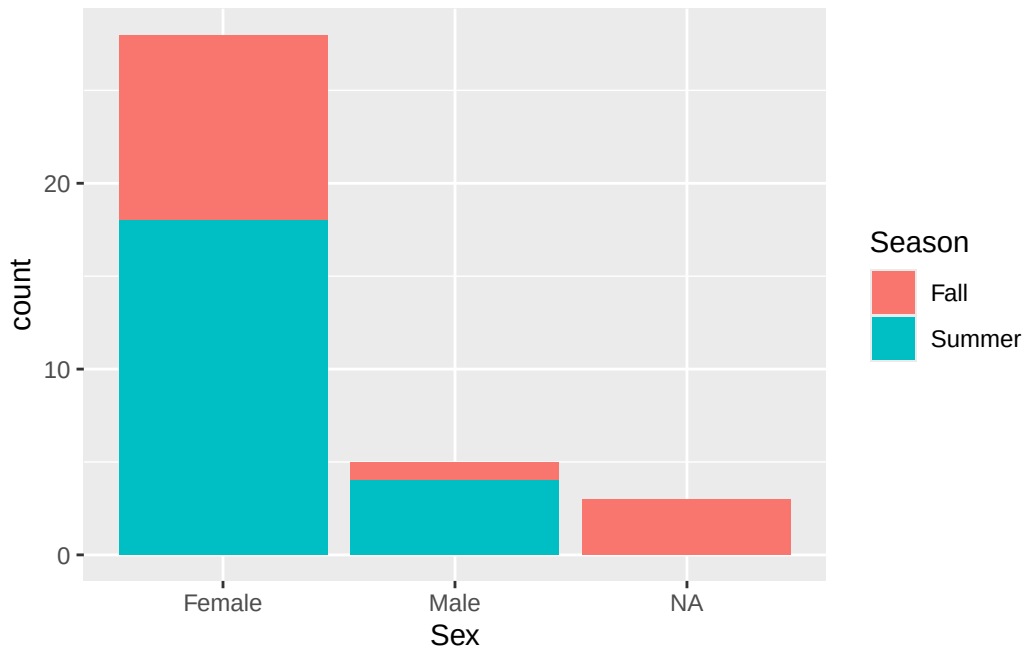


```
#Sexual demographics
table(dfhe$Sex)
```

```
Female  Male
      28    5
```

This map probably doesn't mean much due to the low sample size of males.

```
#Create a bar plot, separating by season and sex
dfhe %>%
  ggplot(aes(x= Sex, fill=Season))+
  geom_bar()
```



There's way more females than males, I'm not sure we can say much from this.

Dipodomys venustus

Of the venustus observations:

F: 19, M: 1, Unknown: 1

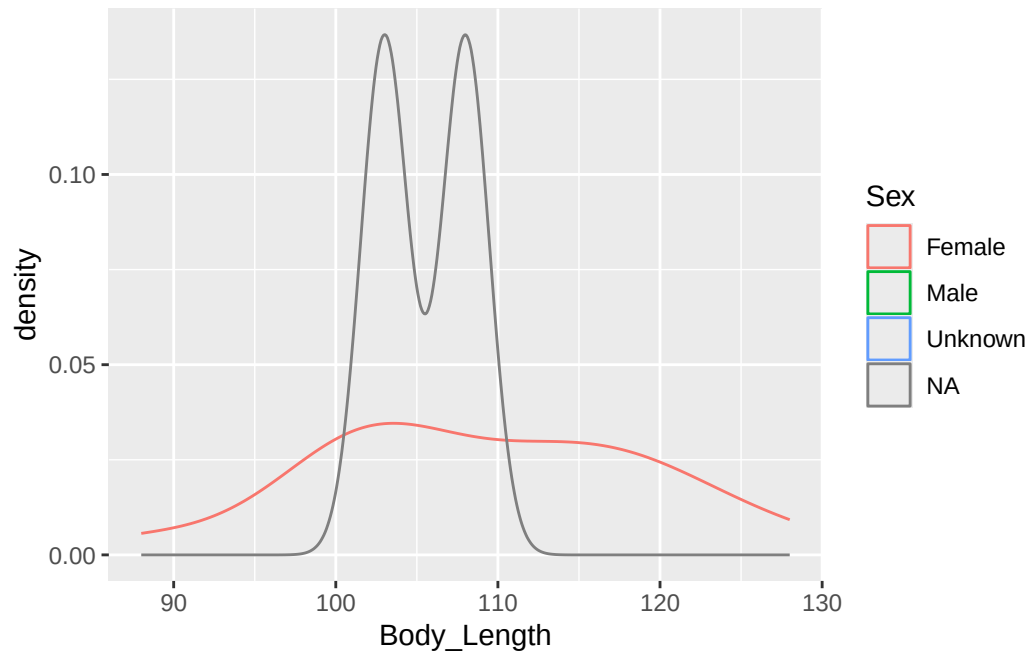
Graph doesn't even work :C

```
#Create a density, smooth line frequency distribution, separating by sex of the venustus and
dfve %>%
  ggplot(aes(x= Body_Length, color =Sex))+
  geom_density()
```

Warning: Removed 1 row containing non-finite outside the scale range
(`stat\_density()`).

Warning: Groups with fewer than two data points have been dropped.
Groups with fewer than two data points have been dropped.

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom\_density()`).



```
#Sexual demographics  
table(dfve$Sex)
```

Female	Male	Unknown
19	1	1

Graph doesn't even work :C